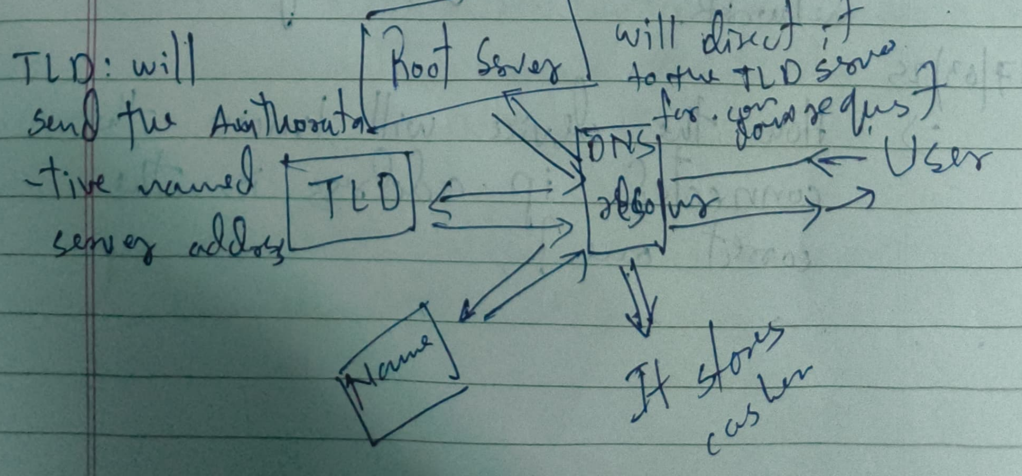
1) IpV4 vs IpV6 — 32 bit & 128 bit long Address
IPV4 was limited address ftr IPV6 was invented.

2) Private & Public IP — ~~Private~~ Public → given by the ISP to direct connect to the internet.
Private → One IP used locally given by the routers.

3) MAC address — Media Access control is the physical address given by the manufacturer to the hardware device assigned to the NIC.

4) DNS — Domain Name System is like a phonebook for Internet which convert the Domain Name to Alpha Numeric IP address.
These do not have physical locations.



DHCP : ~~Domain~~ Dynamic Host Configuration Protocol.

There was a tradition method of assigning IP manually & sub masking which was called static IP. And using DHCP the IP get auto-matically allocated. which is called Dynamic IP.

Subnet : logical division of IP Network.

Dividing large network in smaller Network.

Default Gateway : (Router)

It is a designated device through which the data travels before entering to the group of computer network.

25

=> How the device will know the connected ip address is the correct one.?

Page No.
Date:

192.168.0.2
Network address & Host address

A subnet mask reveals the number of bits in the IP address is Network & Host.
& controles the trafic, security and devides the IP networks to multiple smaller devices.

192.168.0.2 11000000.10101000.00000000.00
255.255.255.0 11111111.11111111.11111.00

Net H H H

192. 168.0.2

& last number is ~~MAC~~
Unique Address through which the correct device is identified.

first 3 octates are checked to verify the same

NAT: Network address translation
It converts Public IP to privet IP & vic versa

router helped in shortage of public IP.

DHCP keeps track of the IP's which are being used

In Dynamic IP configration we use DHCP

<u>Modem</u>: ~~does~~ Connects the ISP to the devises & does the following

Modulation: Converts digital signals to Analog

Demodulation: Vic Versa

Using telephone or wire cable.

<u>Router</u>: Sends the data ~~to~~ b/w multiple devices using IP address

Network layer 3

Routs traffic b/w LAN & WAN

Subnetting: Dividing A big group of Network to small network to manage the data transfer.

Transmission control protocol.
TCP : ~~The~~ TCP is used to mak sure the communication b/w two computer is good and the data recived is complete and correct.

It is a connection oriented protocol so it first confirms or accknowledge a session. b/w the two computer. By A three way ~~ba~~ handshake.

1. The computer sends SYN
2. The ~~sender~~ ~~reciver~~ sender computer sends ~~SYN ASK~~

SYN ACK recieved

TCP makes sure the packets are fully arrived and in case its not reached it will retransfer it.

PORT: Ports are logical connections that are used by the program ~~to~~ and services to exchange information.

NetSTAT : Is used to display the current network connections and port activity.

~~215~~ 215.114.85.17 : 80

Is the port number for HTTP

Vlan: Virtual Lan is a logical Segementation of a physical network or switch net. within the device to act as the distict device

Protocols

HTTP: This protocol is used to view webpages

SSL & TSL is used to encrypt data

Firewall: Is ~~used~~ a system that is designed to prevent unautherised access from entering a private network.

OSI - Model Operating System Interconnected

Application
Presentation
Session
Transport
Network
Data link
Physical

In application layer the protocols work like HTTP's

Application layer:

Protocols: HTTP, HTTPS, FTP, NFS
FMTP, DHCP, SNMP, TELNET
POP_3, IRC, NNTP

FTP → File transfer
HTTPS → Browsing
emailing → SMTP

virtual netw → Telnet

↓ These Data are in form numbers & char

Presentation layer

Translation // PL converts the characters & numbers to Binary form 1110011000l
&
Before transfering the data to next layer it compresses the data and it may also be lossy & lossless

Compression helps to send file in less time.

SSL & TSL → Protocall is used in Presentation layer for encryption & Decryption.

Translation, Encryption & Decription

Session layer:

- API's : NETBIOS (eg)

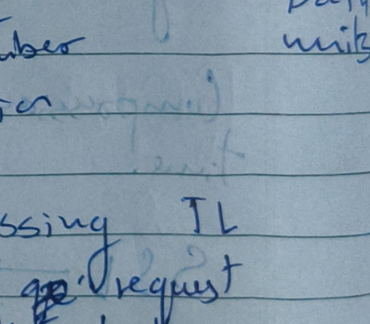
1. Manages the session for data exchange or Transfer.
2. Autherization & Authentication

Transport layer:

TL controles the reliability of Communica[tion] Through

- Segmentation
- Flow control
- Error control

Each segment contains sequence numbers & port number



No. of data transmission

If data segment is missing TL uses Automatic repeat request schemes to retransmit the lost or currupt data

Group of bits called checksum is added to the segments to check if data is corrupted or not

$\boxed{\text{Flow control / Error control}}$ done by

- Connection oriented transmission — FTP, www — TCP
- Connection less — UDP — User Datagram Protocol

TFTP, DNS

No feedback ~~No feed~~

Trasport layer passes The data into segments.

Network layer

- Data units in this layer are Packates
- Routers resides

Functions Are
- ① Logical Addressing : IPv4 & IPv6 + Mask
- Path determination
- Routing

① IP addressing : NL assigns senders and reivers IP's address

OSPF (open shortest path first)
BGP (Border Gateway Protocol)
IS-IS (intermidiate system - IS)

Data packets with IP address

Data link layer

logical addressing has done in Nework layer

Physical addressing is done in Data link layer

Mac Address of sender and reciever are assigned to the packets

MAC1 | MAC2 | IP1 IP2 | tail

Data link layer provide access to media

Physical layer recieves the data in the form of 0's & 1's

It converts

7. Application layer
6. Presentation layer
5. Session layer

} Browser Manages it

session ⟶ cookies & cashe

→ When the request is made first it desides what is the type of request.

L7 ⟶ HTTP, HTTP3, FTP, SMTP, DNS
L6 ⟶ SSL/TLS, JPEG, ASCII, MP3.
L5 ⟶ NETBIOS, PPTP, RPC
L4 ⟶ TCP, UDP, SCTP
L3 ⟶ IP, ICMP, ARP, OSPF, BGP
L2 ⟶ Ethernet, PPL, HDLC, Switch, Bridg
MAC
L1 ⟶ DSL, Wi-fi (radio), Cable. USB

(2) cloud Data Engineer → CP → SA → Data Engineer Associate → Security speciality (deep dive)

(3) Software developent Engineer → cloud pract → SA → AWS certified Devops Engineer (Proffesional)

(4) System Administrator (No code) → CP → SysOps Administrator Associate → Devops Engineer professional

(5) Dev-Sec Ops Engineer → CP → SysOps Admin → Devop Engineer professional

For security Asp → Security speciality

(6) Cloud Security Engineer → CP → Sysops Admin Associate → Security speciality

(7) Network Engineer → CP → Solution Architect Assoc → Advance Networking speciality